EGA: Towards Euclidean Geodesic Alignment for Vector Search on Manifolds

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Abstract

Representation learning and vector search are treated as independent problems, yet they share a fundamental geometric contract that neither community explicitly enforces. Foundation models such as CLIP optimize for semantic alignment on hyperspheres, while downstream search engines assume well-behaved Euclidean partitions, an assumption that latent manifolds systematically violate. We argue this is not an engineering inconvenience but a structural flaw in how the field is organized: retrieval geometry is an afterthought, patched post-hoc through increased search budgets rather than addressed at the source. The community has accepted a 10x computational tax on inference as normal, when it is in fact the cost of ignoring geometry during pretraining. To ground this position empirically, we introduce Euclidean Geodesic Alignment (EGA), a lightweight post-hoc transformation that explicitly flattens local manifold structure. Despite its simplicity, EGA at nprobe=1 matches raw CLIP embeddings at nprobe=5 across CIFAR10 and CIFAR100, demonstrating that geometric correction yields disproportionate retrieval gains. We present this not as a final solution, but as evidence that the field needs to move from post-hoc adaptation to native geometric co-optimization between representation models and retrieval systems.

1 Introduction

The machine learning community has quietly accepted an inefficiency so pervasive it has become invisible. When a practitioner deploys a CLIP-based image retrieval system and finds recall unsatisfactory, the standard remedy is to increase `nprobe`: probing more Voronoi cells at search time until the recall target is met. This workaround is so routine that it appears in the default configurations of production systems and benchmark tutorials alike. We argue it should not be.

This paper takes the position that retrieval geometry is not a downstream engineering concern but a first-class objective that must be co-optimized with representation learning, and that the field's failure to treat it as such imposes a systematic and unnecessary computational tax on every deployed vector search system.

The root cause is a geometric mismatch that has been designed into the stack by accident. Foundation models such as CLIP are trained with contrastive objectives that optimize semantic alignment on a hypersphere: semantically similar inputs are pulled together, dissimilar ones pushed apart. Separately, approximate nearest neighbor (ANN) search engines such as FAISS partition the embedding space into Voronoi cells and assume that semantic neighbors co-locate within the same cell under Euclidean geometry. Neither side is wrong on its own terms. But the two geometric contracts are incompatible, and no one is responsible for reconciling them.

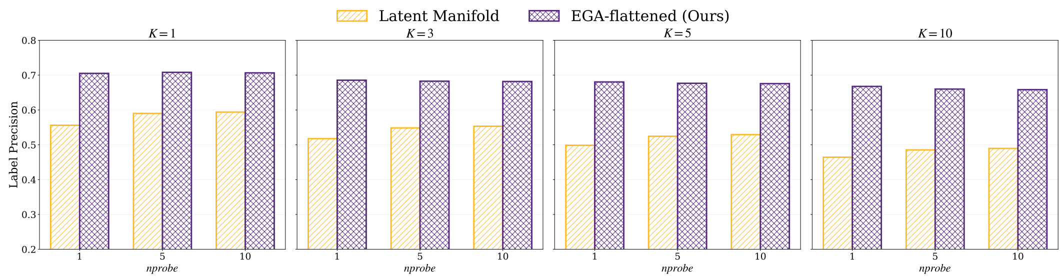

Figure 1: Label precision across varying K and nprobe settings. EGA consistently outperforms the original latent manifold, demonstrating improved semantic alignment.

The consequences are concrete. As we show in Section 4, raw CLIP embeddings on CIFAR100 achieve only 0.51 Recall@1 at `nprobe=1`. Reaching 0.85 recall requires `nprobe=10`: a 10× increase in search cost that buys back accuracy that was never lost in semantic space, only in geometric space. This is not a scaling limitation or a model capacity problem. It is the measurable cost of ignoring geometry at training time.

We introduce Euclidean Geodesic Alignment (EGA) as a proof-of-concept to make this argument empirical rather than theoretical. EGA is a lightweight post-hoc transformation that flattens local manifold structure, enforcing a clear geometric margin between semantic neighbors and background points. It is deliberately simple: our claim is not that EGA is the right solution, but that even a minimal geometric correction applied after the fact yields disproportionate gains: EGA at `nprobe=1` consistently matches or outperforms raw CLIP at `nprobe=5`. If a post-hoc patch this lightweight moves the efficiency frontier this significantly, the gains from native co-optimization should be substantially larger.

The remainder of this paper is organized as follows. Section 2 situates our position within existing work on metric learning, manifold alignment, and retrieval-aware representation. Section 3 formalizes the geometric mismatch and explains why post-hoc corrections are inherently limited. Section 4 presents empirical evidence through EGA. We conclude with a roadmap for how the community should restructure the boundary between representation learning and retrieval systems, in Section 5.

2 Synthesis and Analysis of Literature

3 Theoretical Reasoning

4 Experimental Evidence

4.1 Experimental Setup

Experiment Platform All experiments were conducted on the Chameleon Cloud [3] platform using a compute node equipped with 256 AMD EPYC-7763 CPU cores, 512 GiB of RAM, and an NVIDIA A100 GPU of 80 GB HBM2e. The operating system is Ubuntu 24.04 LTS. [Dongfang: Provide more system setup info.]

4.2 Ground truth of label precision

The metric of label precision is employed to assess the semantic quality of the retrieved results within the approximate nearest neighbor framework. This evaluation focuses on whether the neighbors found in the feature space are categorized under the same physical class as the query. By examining different neighborhood sizes from K=1 to K=10, we can determine the local and global semantic density of the manifold. Our analysis contrasts the original latent manifold with the EGA flattened model to highlight the impact of manifold smoothing on search accuracy.

Figure 1 demonstrates that EGA flattened achieves a substantial improvement in label precision across all configurations. In the K=1 scenario, our method maintains a precision score exceeding

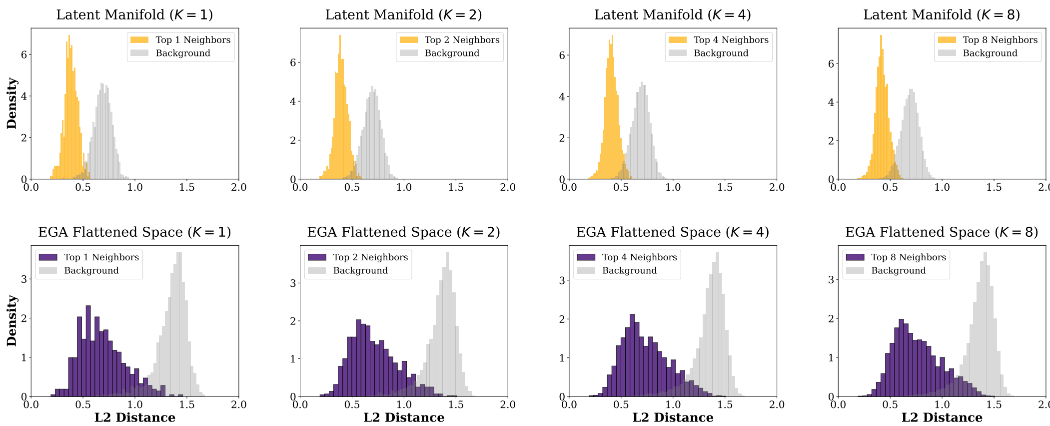

Figure 2: L2 distance distributions for Top K neighbors versus random background points. The original space exhibits significant overlap, whereas EGA enforces a clear geometric margin.

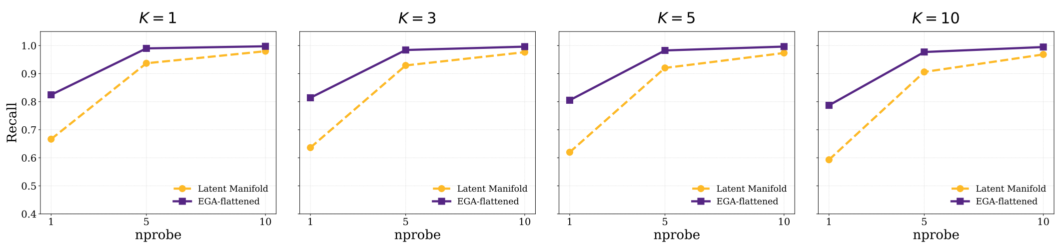

Figure 3: Recall@K versus nprobe tradeoff. EGA consistently shifts the efficiency frontier upward, enabling high recall at minimal search probes.

0.7, while the original latent manifold fluctuates between 0.5 and 0.6. This performance gap remains persistent as the search budget nprobe increases from 1 to 10, indicating that the benefits of our approach are not dependent on extensive search efforts. The superior semantic alignment shown in these figures suggests that EGA creates a more organized and discriminative feature space, allowing the indexer to retrieve correct class labels with higher reliability than the baseline.

4.3 Distances in Latent and EGA Spaces

To validate our stance on the systemic disconnect between semantic alignment and geometric proximity, we present empirical evidence across two dimensions: physical distance distributions and retrieval efficiency metrics. We evaluate Euclidean Geodesic Alignment (EGA) against raw foundational model embeddings using CIFAR10 and CIFAR100 datasets.

The results in Figure 2 expose the root cause of retrieval failure. In the original CLIP latent manifold, the distance distributions of true semantic neighbors and random background points exhibit significant overlap. **This position paper argues that this overlap constitutes a geometric illusion where noise is indistinguishable from signal, rendering Euclidean indexing inherently unreliable.**

4.4 Recalls in Latent and EGA Spaces

The impact of this flattening is quantified in Figure 3. The raw CLIP space achieves a meager 0.51 Recall@1 at nprobe=1 on CIFAR100. To compensate for this disorganized geometry, systems are forced to increase search volume to nprobe=10 to reach a recall of 0.85. This represents a 10x waste of computational resources.

In contrast, EGA elevates the nprobe=1 Recall@1 to 0.68. Notably, EGA at nprobe=1 outperforms the original space at nprobe=5 across all benchmarks. This confirms that fixing the geometry at the source is the only viable path to scalable retrieval. We achieve accuracy parity with brute force multi

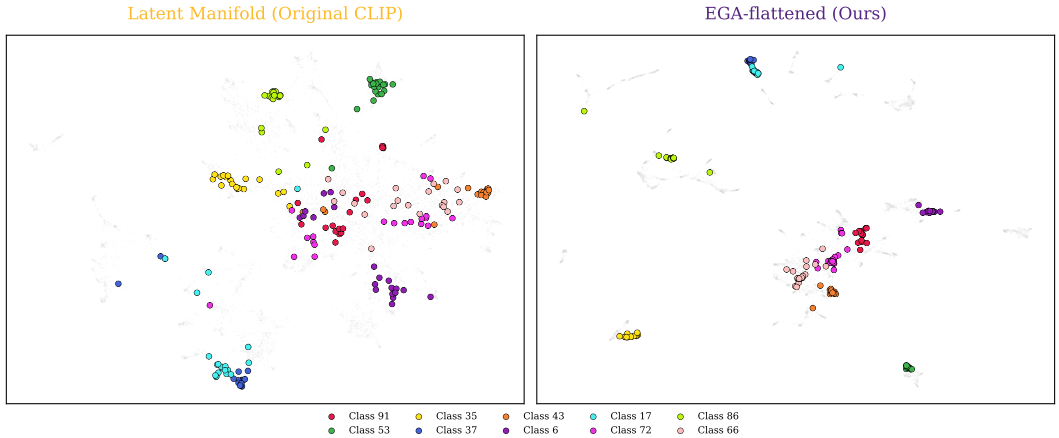

Figure 4: tSNE projection of original CLIP latent manifold (left) versus EGA flattened manifold (right). EGA reveals clearer category clusters and reduced semantic overlap.

probe baselines while maintaining the extreme efficiency of single cell lookups. These results verify that our position is a necessary requirement for the next generation of scalable retrieval systems.

4.5 Low dimensional projection visualization

To better understand the structural changes in the feature space, we utilize t SNE to project high dimensional embeddings into a two dimensional plane. Figure 4 compares the spatial distribution of the latent manifold generated by original CLIP against our proposed method. As observed in the left panel, the baseline manifold is characterized by highly dispersed data points and substantial semantic overlap between different categories. For instance, classes such as Class 91 and Class 53 appear entangled in certain regions, which inherently limits the effectiveness of search algorithms.

The right panel of the figure displays the EGA flattened manifold, revealing a significant enhancement in geometric order. Data points corresponding to specific categories, including Class 35 and Class 72, are tightly clustered and separated by well defined boundaries. This visual evidence confirms that our smoothing technique effectively eliminates manifold irregularities and reduces local noise. Such a structured feature distribution is a direct consequence of the manifold flattening process, which facilitates faster convergence in indexing and contributes to the superior recall rates discussed in previous sections.

4.6 Out-of-Distribution Generalization

The generalization capabilities of EGA are evaluated on the CIFAR-10 dataset, providing a rigorous out-of-distribution (OOD) test since all models were trained exclusively on CIFAR-100. Figure 5 illustrates the recall performance across varying nprobe values and K settings. EGA consistently maintains a dominant lead over all baselines, including recent state-of-the-art methods like ICon [1] and SRL [2].

At a minimum search budget of nprobe=1, EGA achieves a Recall@1 of 0.858, significantly outperforming SRL (0.831) and ICon (0.788). As shown in the inset zooms, EGA continues to hold a clear advantage even at nprobe=10, where other models begin to saturate. A key observation is the efficiency gain provided by our method: EGA achieves a recall level at nprobe=5 that matches or exceeds the performance of supervised baselines at nprobe=10. This indicates that the manifold smoothing provided by EGA effectively doubles the search efficiency in unseen data environments by creating a more index-friendly geometric structure.

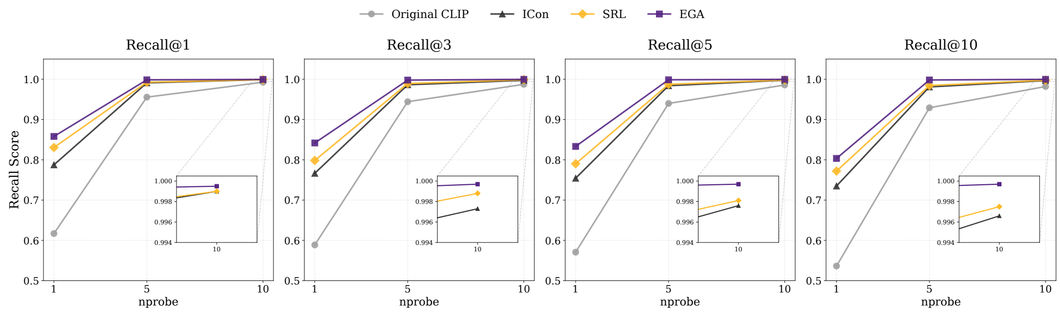

Figure 5: Recall@K versus nprobe tradeoff on CIFAR-10. EGA maintains a consistent lead over baselines, demonstrating strong OOD generalization.

5 Conclusion

References

[1] Shaden Naif Alshammari, John R Hershey, Axel Feldmann, William T Freeman, and Mark Hamilton. I-con: A unifying framework for representation learning. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025.

[2] Zijian Dong, Yilei Wu, Chongyao Chen, Yingtian Zou, Yichi Zhang, and Juan Helen Zhou. Improve representation for imbalanced regression through geometric constraints. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

[3] Kate Keahey, Jason Anderson, Zhuo Zhen, Pierre Riteau, Paul Ruth, Dan Stanzione, Mert Cevik, Jacob Colleran, Haryadi S. Gunawi, Cody Hammock, Joe Mambretti, Alexander Barnes, François Halbach, Alex Rocha, and Joe Stubbs. Lessons learned from the chameleon testbed. In *Proceedings of the 2020 USENIX Annual Technical Conference (USENIX ATC '20)*. USENIX Association, July 2020.